

Human Medians and a Language-Model Forecast: Encompassing and Combination on One ForecastBench Round

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Abstract

I use the one ForecastBench round with individual-level human forecasts (July 2024) to compare two human medians, from 40 superforecaster ids and 500 public forecasters, with the best single language model, chosen on questions no human forecast. The humans did not see the model and could look things up; the model had no news. In preregistered tests the model encompassed neither median (coefficients 1.79 and 0.97), the average superforecaster forecast beat it by 3.4 Brier points $\times 100$, and the average public forecast lost to it by 7.8. In analyses specified after these results, with readings locked before computation, fitted combinations gave the superforecaster median weight 1 and equal-weight pools were worse than that median alone, while the gain from combining the public median with the model on held-out questions had an interval that included zero. The data come from one round, the superforecasters had a group stage, and no news-augmented model variant beat the benchmark, so these results are not evidence about judgment.

Keywords: forecast encompassing; forecast combination; language models; superforecasters; proper scoring rules; ForecastBench

1. Introduction

When two forecasts of the same event differ, a forecaster can ask two questions. Does one forecast contain information the other lacks? And does a combination of the two forecast better on new events? The first is the forecast-encompassing question (Chong & Hendry, 1986; Fair & Shiller, 1990; Harvey, Leybourne & Newbold, 1998). The second is the forecast-combination question, studied since Bates and Granger (1969) and reviewed by Clemen (1989) and Timmermann (2006). Diebold (1989) connected the encompassing and forecast-combination traditions while stressing that in-sample results need not establish the out-of-sample usefulness of a combination. Equal weights often beat estimated weights out of sample, which Claeskens, Magnus, Vasnev and Wang (2016) explain by the noise in estimated weights. For probability forecasts, a combination can average probabilities or log-odds (Genest & Zidek, 1986; Ranjan & Gneiting, 2010), and the verdict can depend on the proper scoring rule used (Gneiting & Raftery, 2007; Merkle & Steyvers, 2013).

Language models now produce probability forecasts on real questions. A retrieval-based system approached the accuracy of a human crowd (Halawi, Zhang, Yueh-Han & Steinhardt, 2024), and an ensemble of twelve language models was statistically indistinguishable from a human crowd on 31 binary questions (Schoenegger et al., 2024). ForecastBench (Karger et al., 2025) scores models and human groups on the same resolving questions. In its 2024 rounds the superforecaster median, from a group selected for forecasting accuracy (Mellers et al., 2014), was ahead of the best models. A ranking of this kind says which forecast is more accurate. It does not say whether the less accurate one adds anything to the other, or whether combining them helps.

This paper asks both questions of human and model forecasts. It uses the one ForecastBench round with individual-level human forecasts, due on 21 July 2024. The model benchmark, M1, is the single model

variant with the best accuracy on questions no human forecast. The humans did not see M1. The study is therefore not about people adjusting a model's forecast. It is about two forecasts made separately for the same events. Section 5 reports the encompassing test in both directions. Section 6 fits combinations on some questions and scores them on others.

The two questions give different answers. M1 encompasses neither the superforecaster median nor the public median. Out of sample, only the superforecaster median beats M1, and there the fitted combination gives M1 no weight: equal-weight averaging with M1 makes the median worse. Given the superforecaster median, M1 is redundant. The public median and M1 do not encompass each other in sample, yet no combination of the two beats M1 reliably on held-out questions. The humans could look things up and M1 had no news, and no news-augmented variant in the round beat M1 (Section 7). The results describe this round and this information gap, not human judgment as such.

2. Data

2.1 Source and round

ForecastBench publishes human forecasts, model forecasts, and resolutions under CC BY-SA 4.0. The human forecasts come from the forecastbench-datasets repository at commit 68932db. The per-model forecasts are not in that repository. They are published as two archives on forecastbench.org, regenerated nightly and unversioned. I fixed one snapshot on 2026-09-06 and recorded its SHA-256 hashes and HTTP ETags in a provenance file. Nothing was re-downloaded during the study.

Only one round, with forecast due date 2024-07-21, contains individual-level human forecasts. It has 500 public forecasters and 40 superforecaster ids on 189 questions. Karger et al. (2025, p. 7) report recruiting 39 superforecasters. In the individual file, one id has 4 forecasts, all on market questions; the next smallest has 63. That id changes the superforecaster median on 2 of the 578 targets. The file has no timestamp or stage field, so I cannot tell which definition gives 39. All 40 ids are kept.

No human forecaster saw a model forecast for these questions. ForecastBench produced the question sets ten days before the forecast due date so that the human surveys could run in that window; the models received their question set 24 hours before the due date (Karger et al., 2025, p. 5). The survey page for a market question showed the question, its resolution criteria, a link to the market, the market's last recorded value at the freeze date, and the freeze date. It showed no model forecast and no supplied news (Karger et al., 2025, Figure 2). Superforecasters recorded the search terms they used and the URLs they consulted (p. 7). The paper describes no limit on looking things up for either group. On market questions this includes the live market price after the freeze; on dataset questions it includes newer data points.

The public forecasters were recruited through Prolific and Facebook advertisements. Each answered 20 random questions from the 200-question human set in a one-hour survey (p. 7). The superforecasters took part in a nine-day exercise. Each was assigned 20 random questions to forecast individually. About halfway through, they moved to a group stage in which they could see one another's forecasts and rationales, communicate about each question, and forecast on questions beyond their assigned 20 (p. 7). This last option is how one id came to forecast all 189 questions.

The unit of observation is a target: a question at one resolution date. The frame keeps targets with a resolution in $\{0, 1\}$: 578 targets on 162 questions, 521 from dataset sources (105 questions) and 57 from prediction markets (57 questions). Each target has at least 3 superforecaster forecasts (median 7) and at least 40 public forecasts (median 49). Superforecasters forecast a median of 87 targets each (range 4 to 578); public forecasters a median of 58 (range 22 to 84).¹

¹One superforecaster id forecast all 578 targets and so enters every superforecaster median. Leaving it out changes the median on 316 targets. The encompassing coefficient for the superforecaster median moves from 1.79 [1.04, 2.53] to 1.68 [0.94, 2.42]; the log-score difference from M1 moves from +0.097 to +0.096; the fitted weight on the median stays at 1 in every fold (Section 6).

Targets nest in questions, and targets of one question at different horizons often share an outcome. The effective number of independent clusters is therefore about 162. Every interval in this paper clusters by question.

2.2 Model forecasts and the matched condition

ForecastBench ran each model in several conditions. The condition that matches the human survey has freeze values and no news. Two prompt scaffolds (zero-shot and scratchpad) ran in that condition for 17 base models, giving 34 variants. Every human target has a forecast from every one of them. ForecastBench fills a model's missing forecasts with 0.5 and flags the row. Flagged rows are excluded from benchmark selection and from the test set, and a model with more than 5% flagged rows on the selection set is ineligible as benchmark. One of the 34 variants was ineligible on that rule. Combination questions, which join two base questions by AND or OR, are excluded everywhere; the human survey contains none.

ForecastBench also ran news-augmented variants in this round. They are not part of the matched condition and enter only in Section 7.

2.3 Data defects

Two hundred public-forecaster rows (0.6%, from 42 forecasters) carried a forecast outside $[0, 1]$, reaching 5.0×10^8 . The public survey asked for a number from 0 to 100, and the published file divides by 100 without cleaning the free-entry field. These rows are not probabilities and are excluded from every analysis. Clipping them to $[0, 1]$ instead moves the public mean Brier difference from -7.81 to -8.09 ($\times 100$). Separately, 242 human market rows could not be mapped to a resolution date and 21,596 rows belonged to unresolved targets. Both are dropped by the resolution filter.

2.4 Specifications

SPEC v1 was locked on 2026-09-06 and pushed as the root commit of the public repository before any analysis. It preregistered the accuracy test (H1, Section 4) and the encompassing test (H2, Section 5). SPEC v2 was locked on 2026-09-08 before its analysis; this paper uses its reference ensemble Mens and its cross-validation folds. SPEC v3 was written after the v1 and v2 results and locked before any of its computations, with the reading of each result fixed in advance. It covers the split by source (Section 5.3), the out-of-sample combination (Section 6), and the news-augmented variants (Section 7). The reverse reading of the encompassing fit (Table 2) and the leave-one-id-out footnote were computed after Section 6. All amendments, in order, and every report the numbers here are drawn from are in the repository.

The analysis code in the repository was generated by Claude (Anthropic; Claude Fable 5.1, accessed through Claude Code, September and October 2026) from the locked specifications and their amendments. I reviewed the code, ran it, and checked every reported number against the repository reports. A notebook in the repository reruns the pipeline from the archived data and checks every number in this paper against the recomputed value.

3. Choosing the benchmark

The benchmark M1 is the single model variant with the best mean Brier score on the selection set: the round's resolved single-question targets minus every target any human forecast. That set has 930 targets on 241 questions and shares no target with the 578 test targets. Selecting on the complement keeps the choice out of sample. Selecting on the human targets would favor whichever model happened to do well on those outcomes and push the human gain down.

Computed after the main analyses.

The winner among the 33 eligible variants was Claude-3-5-Sonnet-20240620 in the zero-shot scaffold with freeze values (selection-set Brier $\times 100 = 16.16$), ahead of the same model's scratchpad variant (16.34). The restriction to single questions mattered. On the unrestricted set, where 86% of targets are combination questions, a different model would have won.

A single benchmark can be lucky or unlucky on the test set. Section 6 therefore also uses Mens, the median of the 33 matched variants other than M1 on each target.

4. Accuracy

Accuracy is reported at two levels, and the two give different answers for the public.

The preregistered test (H1) works at the level of individual forecasts. For each forecaster and target, the gain is the Brier score of M1 minus the Brier score of the human forecast, so a positive gain means the human was closer to the outcome. The mean over 33,334 forecaster-target rows, with question-clustered intervals, is $+3.38 \times 100$ [1.36, 5.41] for superforecasters and -7.81×100 [-9.69, -5.92] for public forecasters. The typical superforecaster forecast beat M1; the typical public forecast lost to it.

Sections 5 and 6 work with the median forecast of each group on each target. At that level the comparison changes for the public (Table 1). The superforecaster median beats M1 in log score by 0.097 [0.060, 0.140] and in Brier score by 0.035 [0.020, 0.052]. The public median does not differ from M1: -0.018 [-0.060, +0.027] in log score and -0.001 [-0.019, +0.017] in Brier score. Taking the median of about 49 public forecasts removes most of what the individual forecasts lose to M1. These median comparisons involve no fitting. M1 was chosen on other questions, and both forecasts are scored on the same 578 targets. They were computed after the main results.

Table 1. Accuracy against M1. Positive values favor the human forecast. Per-forecast rows: mean over forecaster-target rows, Brier $\times 100$, preregistered (H1). Median rows: one forecast per target, raw units, computed after the main results. Question-clustered 95% intervals.

Level	Group	Brier	Log score
Per forecast ($\times 100$)	Superforecasters	+3.38 [1.36, 5.41]	–
Per forecast ($\times 100$)	Public	-7.81 [-9.69, -5.92]	–
Median (raw)	Superforecasters	+0.035 [+0.020, +0.052]	+0.097 [+0.060, +0.140]
Median (raw)	Public	-0.001 [-0.019, +0.017]	-0.018 [-0.060, +0.027]

– : no preregistered per-forecast log-score test.

Two facts limit what the superforecaster result can mean. Superforecasters had a group stage in which they saw one another's forecasts and rationales, and M1 forecast once, alone. Both human groups had access to information M1 did not have (Section 2.1). A positive result for superforecasters is a statement about a deliberating, searching group against a single model without news. It is not a measure of judgment alone.

5. Encompassing

5.1 The test

The outcome is fixed within a target, so the human forecasts are aggregated first: the median superforecaster forecast on the target, and separately the median public forecast. Write p_a for the M1 forecast and p_h for a human median. The preregistered test (H2) is the forecast-encompassing regression of Chong and Hendry (1986), as a logistic model with question-clustered standard errors:

$$\text{logit } P(o = 1) = c + b_a \cdot \text{logit}(p_a) + b_h \cdot [\text{logit}(p_h) - \text{logit}(p_a)].$$

If $b_h > 0$, M1 does not encompass the human median: the median carries outcome information that M1 lacks. Probabilities are clipped to [0.01, 0.99] before the logit. The same fitted model can be written as

$$\text{logit } P(o = 1) = c + (b_a - b_h) \cdot \text{logit}(p_a) + b_h \cdot \text{logit}(p_h).$$

So $b_a - b_h$ is the coefficient on M1 given the human median. If it is zero, the median encompasses M1. This reverse reading uses no new model and no new data. Its interval comes from the question-clustered covariance of the same fit. It was computed after the out-of-sample combination in Section 6, and Table 2 labels it that way.

5.2 Results

Table 2. Encompassing in both directions. Logistic regression of the outcome on M1 and a human median; 578 targets, 162 question clusters; clip [0.01, 0.99]. Column 2 is preregistered (H2). Column 3 is the same fit read in reverse, computed after the analyses in Section 6. 95% intervals.

Human median	M1 does not encompass the median: b_h	The median does not encompass M1: $b_a - b_h$
Superforecasters	1.79 [1.04, 2.53]	-0.23 [-0.77, 0.30]
Public	0.97 [0.24, 1.70]	0.72 [0.32, 1.12]

M1 encompasses neither median. Both b_h intervals lie above zero.

The reverse direction separates the groups. For the superforecaster median, the coefficient on M1 is negative and its interval includes zero: the data do not reject that the superforecaster median encompasses M1. For the public median, the coefficient on M1 is positive and its interval excludes zero. Neither the public median nor M1 encompasses the other in sample. Section 6 asks whether either result turns into a gain when combination weights are fitted on other questions.

The superforecaster median hit the clip bound on 160 of 578 targets; the public median hit it on none. With bounds of [0.001, 0.999], b_h is 1.76 [0.99, 2.54] for superforecasters and 0.97 [0.24, 1.70] for the public, and the reverse coefficients are -0.24 [-0.77, 0.29] for superforecasters and 0.72 [0.32, 1.12] for the public. Reverse coefficients by source are in Appendix Table A4 and are not interpreted.

5.3 By source

The 578 targets come from two kinds of source, and the information gap between humans and M1 differs between them. On dataset questions the humans could look up newer data points and M1 could not. On market questions both had the market value at the freeze date. A specification written after the main results and locked before computation split the test by source. It read a positive coefficient on both subsets as evidence that the human increment is not fully explained by newer data points. The coefficients were positive on both: on dataset targets 2.16 [1.04, 3.28] for superforecasters and 1.08 [0.26, 1.90] for the public; on market targets 1.12 [0.13, 2.11] and 1.94 [0.34, 3.55] (57 targets). The premise of that reading does not hold. The humans could still see the market price after the freeze, so on market questions the information gap is smaller, not absent. The split therefore does not rule out the information-set account, and it is not evidence of anything beyond the freeze price. The coefficients are not compared across subsets.

6. Out-of-sample combination

Section 5 tested encompassing in sample. A forecaster who wants to use the human median needs a different answer: does a combination with weights fitted on other questions score better on new ones? This section reports that test. It was specified after the main results and locked before computation.

6.1 Design

The 162 questions are split into five folds, the same folds as the cross-fitted analysis in SPEC v2. Weights are fitted on four folds and scored on the fifth, and the scores are pooled over the five test folds. The sources are M1, the two human medians, and Mens, the median of the 33 matched variants other than M1. Four combiners were fixed in advance: C1, the equal-weight average of probabilities; C2, the equal-weight average of log-odds; C3, a linear pool with one weight in $[0, 1]$ chosen to minimize Brier on the training folds; C4, a log-odds pool with one weight in $[0, 1]$ chosen to minimize the log score. The log score is primary, with probabilities clipped to $[0.01, 0.99]$; the Brier score is reported beside it. Intervals come from a question-level cluster bootstrap with 2,000 draws, in which folds are reassigned and C3 and C4 weights are refitted inside each draw.

The locked success criterion was that human + M1 beats M1 alone and also beats M1 + Mens, with both intervals above zero. The second condition was meant to rule out a gain that any second forecast would give. It turned out to be the weaker condition: Mens alone scores worse than M1 (log -0.511 against -0.448), so M1 + Mens is worse than M1 too. The binding condition is beating M1. A second locked rule makes C3 and C4 descriptive only if a fitted weight varies across folds with SD above 0.2. No weight came close; the largest fold SD was 0.099, for the public median with M1 under C3.

6.2 Results

Table 3. Out-of-sample scores, pooled over five question folds. 578 targets, 162 questions. Log score: mean log probability of the outcome (higher is better). Brier: mean squared error (lower is better). Specified after the main results; readings locked before computation. Panel A: scores by system.

System	Log score	Brier
M1	-0.448	0.153
Mens	-0.511	0.173
Superforecaster median	-0.351	0.118
Public median	-0.466	0.154
Superforecaster median + M1, equal weights (C1)	-0.385	0.129
Superforecaster median + M1, fitted weights (C3)	-0.351	0.118
Public median + M1, equal weights (C1)	-0.440	0.146
Public median + M1, fitted weights (C3)	-0.444	0.148
M1 + Mens, equal weights (C1)	-0.470	0.158
Superforecaster median + Mens, equal weights (C1)	-0.416	0.134
Public median + Mens, equal weights (C1)	-0.473	0.156

Table 3, Panel B. Differences (first minus second; positive favors the first), with 95% intervals.

Comparison	Log score	Brier
Superforecaster median vs M1	+0.097 [+0.060, +0.140]	+0.035 [+0.020, +0.052]
Superforecaster median vs its equal-weight pool with M1 (C1)	+0.034 [+0.020, +0.050]	+0.011 [+0.005, +0.017]

Comparison	Log score	Brier
Superforecaster median vs its equal-weight pool with M1 (C2)	+0.025 [+0.012, +0.039]	+0.008 [+0.003, +0.014]
Public median vs M1	-0.018 [-0.060, +0.027]	-0.001 [-0.019, +0.017]
Public median + M1 (C1) vs M1	+0.008 [-0.014, +0.034]	+0.007 [-0.002, +0.016]
Public median + M1 (C3) vs M1	+0.004 [-0.007, +0.031]	+0.005 [-0.001, +0.017]
Mens vs M1	-0.063 [-0.094, -0.032]	-0.020 [-0.032, -0.008]

Scores for C2 and C4 and the full set of locked comparisons are in Appendix Tables A1 and A2.

Superforecaster median. Under both fitted combiners the superforecaster median receives weight 1 in every fold. The fitted pool is therefore the superforecaster median itself. The equal-weight pools are worse than the median alone, by 0.034 (C1) and 0.025 (C2) in log score, with intervals above zero. Given the superforecaster median, M1 is redundant; it is not a component of a better mix. This matches the reverse encompassing result in Section 5, where the coefficient on M1 given the superforecaster median was $-0.23 [-0.77, 0.30]$. When the weights are left unconstrained (a reading of the specification chosen after it was written), the fit puts 1.24 on the median and -0.17 on M1, again pointing away from the model. The locked success criterion is met under all four combiners, but at a corner solution: it is met because the median alone beats M1.

Claeskens, Magnus, Vasnev and Wang (2016) explain why equal weights often beat estimated weights: when the true weights are close to equal, estimation noise costs more than the weights gain. The case here is the opposite. The weight sits at the boundary with no variation across folds, and estimated weights beat equal weights.

Public median. In sample, the public median and M1 do not encompass each other (Section 5). Out of sample, no combination with M1 is reliably better than M1. The equal-weight pool improves on M1 by 0.008 in log score, with an interval from -0.014 to $+0.034$; the fitted pool by 0.004, from -0.007 to $+0.031$. The other combiners give the same picture. Unconstrained weights are close to equal (0.52 on the median, 0.50 on M1). The public median scores about the same as M1 (log -0.018 , Brier -0.001 , both intervals including zero). So the two forecasts are of similar accuracy, each carries some information the other lacks in sample, and averaging them still does not produce a gain that holds on new questions. Rejecting encompassing in sample is not the same as a deployable combination gain.

6.3 Against the model ensemble

The locked co-primary question was whether each human median adds to Mens. Both do, under all four combiners: the superforecaster median by $+0.095$ to $+0.161$ in log score and the public median by $+0.038$ to $+0.044$, all intervals above zero. Because Mens is weaker than M1, this is a low bar.

A noise control asks whether the superforecaster result is only a matter of sample size: superforecaster medians pool about 7 forecasts per target, public medians about 49. Replacing the public median by the median of 7 public forecasts drawn at random on each target, over 200 draws, gives an improvement over Mens between about $+0.010$ and $+0.043$ depending on the combiner (2.5th to 97.5th percentile). The superforecaster improvement lies far above that range under every combiner. The full public median lies close to its upper end. The superforecaster result is therefore not explained by the smaller number of forecasts in its median. It is not thereby explained by skill either: the group stage and the selection of superforecasters both sit inside that label. The criterion for "behaves like the superforecaster median" was set after the specification was written.

The locked three-source pools used C1 and C2 only. Consistent with the redundancy in Section 6.2, under C2 adding M1 to the superforecaster-median-plus-Mens combination lowers the log score by 0.011

[0.003, 0.020]; under C1 the change is a decrease of 0.004 with an interval that includes zero ([−0.003, 0.012]).

7. The information set

Every result above compares human forecasters who could look things up with a model that had no news. Part of the human advantage may be retrievable information rather than judgment. ForecastBench also ran news-augmented variants in this round, in which retrieved news articles were summarized and added to the prompt. A specification written after the main results and locked before computation asked whether any of them could narrow this gap. Its reading was fixed in advance: if no news variant scored better than M1 on the selection set, the comparison could not narrow the gap and the human coefficients against the news variants would not be interpreted.

No news variant scored better than M1. On the selection set, M1 has $\text{Brier} \times 100 = 16.16$. The best news variant of any base model was Claude-3-5-Sonnet-20240620 with the first superforecaster-written news prompt, at 17.37. Its prompt was written by superforecasters for ForecastBench's LLM ensemble (Karger et al., 2025, p. 8). It contains the question and retrieved news summaries and no human forecast. The two Sonnet scratchpad variants with news scored 17.94 and 17.63. On the test targets the best news variant was also no better than M1 (log score -0.021 [−0.069, +0.020]; Brier -0.006 [−0.026, +0.010]).

So the ForecastBench news pipeline did not improve this model in this round, and this round cannot separate retrievable information from judgment. The encompassing coefficients against each news variant are reported in Appendix Table A3 and are not interpreted. Every claim in this paper is relative to a model without news.

8. Discussion and limits

In this round the superforecaster median beat the best single model and made it redundant. No fitted combination gave the model weight, and averaging the two lost accuracy. The public median tied the model. In sample, each carried information the other lacked; out of sample, averaging them did not produce a gain that held on new questions.

The two results come from asking the encompassing question and the combination question separately. The in-sample encompassing test rejected for both medians. Someone who stopped there would combine both medians with the model. The cross-fitted test separates them: the public median does not reliably improve the model, and the superforecaster median should replace it rather than be averaged with it. With 162 question clusters, an in-sample rejection is a weak guide to whether estimated combination weights will help on new questions.

The results do not separate judgment from information. The humans could look things up, and the superforecasters deliberated in a group stage. M1 had no news, and the news-augmented variants in the round were less accurate than M1, so this round cannot say how much of the human advantage a better retrieval system would remove. The superforecaster result is a result about a selected group working under tournament conditions with web access, not about people in general.

Limits. The data come from one round, the only one with individual-level human forecasts, so nothing here is checked across rounds. The effective sample is about 162 questions, and the market subset has 57 targets. The superforecaster median pools about 7 forecasts per target; leaving out the one forecaster present on every target does not change any conclusion (footnote, Section 2.1). M1 is a mid-2024 model. Later model generations may change the comparison, and this round cannot say how. The out-of-sample combination and the news comparison were specified after the main results, with readings locked before

computation; the reverse encompassing reading and the leave-one-id-out check were computed after them.

Two questions this paper sets aside are treated in a companion methods paper (Sun, in preparation): how per-forecaster gain over a model depends on which model serves as the reference, and how conclusions about extremizing relative to a model depend on the scoring rule.

In one ForecastBench round, the best single language model without news encompassed neither human median. Only the superforecaster median produced a gain over the model on held-out questions, and the fitted combination gave the model no weight.

Data and code availability

The three specifications with all their amendments, the analysis code, the reports for every number in this paper, and a notebook that reruns the pipeline and checks each reported number against the recomputed value are in the public repository github.com/aaronsun923/forecast-complementarity, archived at <https://doi.org/10.5281/zenodo.22600832>. The ForecastBench human forecasts are in the publisher's public datasets repository at commit 68932db. The publisher regenerates its model forecast sets nightly without versioning; the unmodified snapshot of 2026-09-06 used here is archived under its CC BY-SA 4.0 licence at <https://doi.org/10.5281/zenodo.23201267>. A preprint of this paper is archived at <https://doi.org/10.5281/zenodo.22589676>.

Author note

The author designed the study, wrote and locked the specifications, verified the results, and wrote the manuscript. The author thanks the ForecastBench team for publishing the individual-level forecasts this study depends on. The author declares no conflicts of interest and received no funding for this work.

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Appendix

Table A1. Out-of-sample scores for every two-source pool and combiner, and the locked three-source pools. Pooled over five question folds; 578 targets. Log score higher is better; Brier lower is better.

Pool	Combiner	Log score	Brier
Superforecaster median + M1	C1	-0.385	0.129
Superforecaster median + M1	C2	-0.375	0.127
Superforecaster median + M1	C3	-0.351	0.118
Superforecaster median + M1	C4	-0.351	0.118
Public median + M1	C1	-0.440	0.146
Public median + M1	C2	-0.435	0.146
Public median + M1	C3	-0.444	0.148
Public median + M1	C4	-0.439	0.148
Superforecaster median + Mens	C1	-0.416	0.134
Superforecaster median + Mens	C2	-0.385	0.127
Superforecaster median + Mens	C3	-0.351	0.118
Superforecaster median + Mens	C4	-0.351	0.118
Public median + Mens	C1	-0.473	0.156
Public median + Mens	C2	-0.469	0.154
Public median + Mens	C3	-0.470	0.155
Public median + Mens	C4	-0.467	0.155
M1 + Mens	C1	-0.470	0.158
M1 + Mens	C2	-0.461	0.156
M1 + Mens	C3	-0.451	0.154
M1 + Mens	C4	-0.449	0.154
Superforecaster median + M1 + Mens	C1	-0.420	0.138
Superforecaster median + M1 + Mens	C2	-0.396	0.132
Public median + M1 + Mens	C1	-0.457	0.151
Public median + M1 + Mens	C2	-0.448	0.149

Single sources: M1 -0.448 / 0.153; Mens -0.511 / 0.173; superforecaster median -0.351 / 0.118; public median -0.466 / 0.154. Unconstrained linear weights (C3u, secondary) are in the repository report.

Table A2. Locked comparisons, log score and Brier differences with 95% question-cluster bootstrap intervals (2,000 draws). Positive favors the first system.

Comparison	Combiner	Log score	Brier
Superforecaster median + M1 vs M1	C1	+0.063 [+0.040, +0.090]	+0.024 [+0.015, +0.035]
Superforecaster median + M1 vs M1 + Mens	C1	+0.085 [+0.066, +0.105]	+0.029 [+0.021, +0.038]
Superforecaster median + M1 vs M1	C2	+0.073 [+0.047, +0.102]	+0.027 [+0.017, +0.038]
Superforecaster median + M1 vs M1 + Mens	C2	+0.086 [+0.064, +0.108]	+0.029 [+0.020, +0.039]
Superforecaster median + M1 vs M1	C3	+0.097 [+0.060, +0.140]	+0.035 [+0.020, +0.052]
Superforecaster median + M1 vs M1 + Mens	C3	+0.100 [+0.060, +0.142]	+0.036 [+0.020, +0.052]
Superforecaster median + M1 vs M1	C4	+0.097 [+0.060, +0.140]	+0.035 [+0.020, +0.052]
Superforecaster median + M1 vs M1 + Mens	C4	+0.099 [+0.060, +0.141]	+0.035 [+0.020, +0.052]

Comparison	Combiner	Log score	Brier
Public median + M1 vs M1	C1	+0.008 [−0.014, +0.034]	+0.007 [−0.002, +0.016]
Public median + M1 vs M1 + Mens	C1	+0.030 [+0.013, +0.048]	+0.012 [+0.004, +0.020]
Public median + M1 vs M1	C2	+0.013 [−0.008, +0.036]	+0.008 [−0.001, +0.017]
Public median + M1 vs M1 + Mens	C2	+0.026 [+0.008, +0.043]	+0.010 [+0.002, +0.018]
Public median + M1 vs M1	C3	+0.004 [−0.007, +0.031]	+0.005 [−0.001, +0.017]
Public median + M1 vs M1 + Mens	C3	+0.007 [−0.006, +0.033]	+0.005 [−0.001, +0.017]
Public median + M1 vs M1	C4	+0.009 [−0.003, +0.035]	+0.005 [−0.000, +0.018]
Public median + M1 vs M1 + Mens	C4	+0.011 [−0.002, +0.036]	+0.006 [−0.000, +0.018]
Superforecaster median + Mens vs Mens	C1	+0.095 [+0.075, +0.115]	+0.039 [+0.029, +0.049]
Superforecaster median + Mens vs Mens	C2	+0.126 [+0.101, +0.153]	+0.047 [+0.036, +0.059]
Superforecaster median + Mens vs Mens	C3	+0.161 [+0.124, +0.197]	+0.055 [+0.040, +0.070]
Superforecaster median + Mens vs Mens	C4	+0.161 [+0.125, +0.197]	+0.055 [+0.040, +0.070]
Public median + Mens vs Mens	C1	+0.038 [+0.019, +0.057]	+0.017 [+0.008, +0.027]
Public median + Mens vs Mens	C2	+0.043 [+0.023, +0.063]	+0.019 [+0.009, +0.029]
Public median + Mens vs Mens	C3	+0.042 [+0.016, +0.083]	+0.018 [+0.007, +0.036]
Public median + Mens vs Mens	C4	+0.044 [+0.018, +0.083]	+0.019 [+0.008, +0.036]
Superforecaster median + M1 + Mens vs M1 + Mens	C1	+0.050 [+0.037, +0.064]	+0.020 [+0.014, +0.027]
Superforecaster median + M1 + Mens vs M1 + Mens	C2	+0.065 [+0.048, +0.082]	+0.024 [+0.017, +0.032]
Public median + M1 + Mens vs M1 + Mens	C1	+0.013 [+0.001, +0.025]	+0.007 [+0.002, +0.013]
Public median + M1 + Mens vs M1 + Mens	C2	+0.013 [+0.001, +0.026]	+0.007 [+0.002, +0.013]
Superforecaster median + M1 + Mens vs superforecaster median + Mens	C1	−0.004 [−0.012, +0.003]	−0.004 [−0.007, −0.000]
Superforecaster median + M1 + Mens vs superforecaster median + Mens	C2	−0.011 [−0.020, −0.003]	−0.005 [−0.009, −0.002]
Public median + M1 + Mens vs public median + Mens	C1	+0.016 [+0.007, +0.025]	+0.005 [+0.000, +0.009]
Public median + M1 + Mens vs public median + Mens	C2	+0.021 [+0.009, +0.032]	+0.006 [+0.000, +0.010]

Table A3. Encompassing coefficients against the news-augmented variants (SPEC v3, Amendment 5). Same regression as Table 2, column 2, with the news variant in place of M1; 578 targets. Difference: coefficient against the news variant minus coefficient against M1, question-cluster bootstrap (2,000 draws). Under the locked reading these coefficients are not interpreted, because no news variant beat M1 on the selection set.

News variant (Claude-3-5-Sonnet-20240620)	Selection Brier ×100	Group	Coefficient	Difference from M1
Superforecaster prompt 1 with news	17.37	Superforecasters	1.48 [0.82, 2.14]	−0.31 [−0.82, 0.05]
		Public	1.11 [0.44, 1.78]	0.14 [−0.31, 0.58]
Scratchpad with news and freeze values	17.63	Superforecasters	1.41 [0.73, 2.10]	−0.37 [−0.76, −0.08]
		Public	0.73 [0.11, 1.35]	−0.24 [−0.58, 0.09]

News variant (Claude-3-5-Sonnet-20240620)	Selection Brier $\times 100$	Group	Coefficient	Difference from M1
Scratchpad with news	17.94	Superforecasters	1.41 [0.74, 2.08]	-0.38 [-0.81, -0.06]
		Public	0.81 [0.22, 1.40]	-0.16 [-0.50, 0.16]
Scratchpad with second news	21.36	Superforecasters	1.67 [0.95, 2.39]	-0.12 [-0.54, 0.28]
		Public	1.51 [0.91, 2.11]	0.54 [0.21, 0.89]

M1 selection-set Brier $\times 100 = 16.16$. The scratchpad-with-second-news variant was not named in the specification and is shown for completeness.

Table A4. Reverse encompassing by source: coefficient on M1 given the human median ($b_a - b_h$), same fit as Table 2, column 3, estimated separately on each subset. Computed after the analyses in Section 6; not interpreted.

Subset	Human median	$b_a - b_h$ [95% CI]	Targets	Questions
Dataset	Superforecasters	0.01 [-0.76, 0.78]	521	105
Dataset	Public	1.04 [0.43, 1.65]	521	105
Market	Superforecasters	-0.30 [-1.20, 0.61]	57	57
Market	Public	-0.61 [-1.45, 0.23]	57	57